National University of Computer and Emerging Sciences
Islamabad, Pakistan

Assignment # 4 – Linear and Logistic Regression

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Course: Artificial Intelligence

Instructions:

- You are allowed to discuss this assignment verbally but plagiarism is not acceptable at all.
- If there are any assignment related queries, first read the document carefully. If your query is still unresolved, you can e-mail your TA or discuss with the instructor.
- Submit your file under the naming **convention iXXXXXX_Section.ipynb (E.g. i210328_A.ipynb).**
- Submit the assignment on time. Late submissions will receive a 10% penalty per day (acceptable for next two late days only).
- Assignment will be due by **1 December 2024, Sunday 11:59 p.m.**
- This assignment is handwritten and code base so no printed format will be acceptable.

Use the given sample dataset and solve the below given questions.

Train Samples:

Age	Income (1000's)	Credit Score	Estimated Loan 1000's (Regression)	Loan Status (Binary Class)	Loan Category (Multi-Label)	Loan Category (Multi-Class)
32	85	720	17	Approved	[1,1,0]	B or [0,1,0]
28	25	590	5	Denied	[0,1,1]	C or [0,0,1]
51	105	780	21	Approved	[1,1,0]	A or [1,0,0]
36	75	710	15	Approved	[0,1,0]	B or [0,1,0]
24	48	640	9.6	Denied	[0,1,1]	C or [0,0,1]

Test Sample:

Age	Income	Credit Score	Estimated Loan (Regression)	Loan Status (Binary Class)	Loan Category (Multi-Label)	Loan Category (Multi-Class)
45	62000	680	**?**	**?**	**?**	**?**
45	62000	680	12400	Approved	[0,1,0]	B or [0,1,0]

Q1: Linear Regression

You have to:

1. Use Income and Credit Score as features to predict the Estimated Loan Price using a Linear Regression Model.
2. Use the given estimated loan price as the ground truth and calculate the Mean Squared Error (MSE):
    - After one epoch of training, using the initial weights.
    - After 10 epochs of training, using the final updated weights.
3. First, perform manual calculations for one epoch (prediction, error calculation, and weights update).
4. Then, implement the model programmatically to train for 10 epochs and calculate the updated MSE using the final weights.

Q2: Logistic Regression

You have to:

1. Use Income and Credit Score as features to classify whether the loan is Approved or Denied using a Logistic Regression Model.
2. Choose a suitable activation function and loss function from the provided cheat sheet.
3. Perform the following:
    - Manual calculations for one epoch: Predict probabilities, calculate the error (loss), and update weights.
    - Train the model programmatically for 10 epochs and use the final weights to recalculate the error.

Q3: Multi-Label Classification

You have to:

1. Use Income and Credit Score as features to predict the loan categories for which a user is eligible.
2. Each user can belong to multiple categories simultaneously.
3. Choose a suitable activation function and loss function from the provided cheat sheet.
4. Perform the following:
    - Manual calculations for one epoch: Predict probabilities, calculate the error (loss), and update weights.
    - Train the model programmatically for 10 epochs and use the final weights to recalculate the error.

Q4: Multi-Class Classification

You have to:

1. Use Income and Credit Score as features to predict the loan category a user belongs to.
    - In this case, a user can belong to only one category at a time.
2. Choose a suitable activation function and loss function from the provided cheat sheet.
3. Perform the following:
    - Manual calculations for one epoch: Predict probabilities, calculate the error (loss), and update weights.
    - Train the model programmatically for 10 epochs and use the final weights to recalculate the error.

Important Note

- Use your NU id's last three digits shifting one decimal point left as the given initial weights. For example 21i-1375 (w0=0.3, w1=0.7,w3=0.5). Also, to make calculation's easier and more manageable we have considered income and estimated loan values in 1000's. For example, 85000 is equal to 85. So remember while predicting on test sample manage it accordingly. You have to give prediction value on test sample in each question. Second row in the test is the actual answer just for your information.
- Start with manual calculations (prediction, error, and weight updates) for one epoch.
- Then, implement programmatically for 10 epochs. I will ask your lab instructor to help you in this if you have any queries feel free to discuss.
- Use the provided cheat sheet for help with error functions, their derivatives, equations, and weight update rules.
- You may use separate equations for each weight or the matrix multiplication method both are given in cheat sheet.
- **Cheat sheet is generated using AI**, so if you found any issues please highlight.

Rubrics for marking:

Question 1	10
Question 2	10
Question 3	10
Question 4	10
Implementation	40
Total	80

Attempt this by heart and you will really enjoy it, anyways you have to do it.

----------***Happy Learning Machine Learning***----------